

Set-1

1. Choose True/False for the following statements.

$1 \times 5 = 5$

- (a) A *strictly dominant* strategy may not be unique. **False**
- (b) When a strategic form game has *multiple* pure strategy Nash equilibria (PSNE), the payoffs for a player can be different in different PSNEs. **True**
- (c) In a pure strategy Nash equilibrium (PSNE), the equilibrium strategy of each player offers a best response (BR) irrespective of the strategies of the rest of the players. **False**
- (d) A game with a finite number of players and strategies has *at least one* Mixed Strategy Nash Equilibrium (MSNE). **True**
- (e) In MSNE, the expected utility of a player i must be same for any strategy that has probability $(0, 1]$. **True**

2. Find PSNE for the following game:

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		P-2			
		A	B	C	D
P-1	A	5, 2	2, 6	1, 4	0, 4
	B	0, 0	3, 2	2, 1	1, 1
	C	7, 0	2, 2	1, 5	5, 1
	D	9, 5	1, 3	0, 2	4, 8

Solution:

2.

	A	B	C	D
A	5, 2	2, 6	1, 4	0, 4
B	0, 0	3, 2	2, 1	1, 1
C	7, 0	2, 2	1, 5	5, 1
D	9, 5	1, 3	0, 2	4, 8

Using the $\square\bigcirc$ method, we have

- $\square$: Best response correspondence of player 1 for a given strategy of player 2.
- $\bigcirc$: Best response correspondence of player 2 for a given strategy of player 1.

As $\square$ and $\bigcirc$ coincide at (B, B) , the PSNE of the game is (B, B) .

Evaluation:

- For correct method and answer: 5
- For partially correct method that shows the student knows the concept: 2.5
- Else: 0

3. Assume two bidders with valuations v_1 and v_2 for an object. Their bids are in multiple of some unit (that is, discrete). The bidder with higher bid wins the auction and pays the amount that he has bid. If both bid the same amount, one of them gets the object with equal probability 0.5. Find a PSNE of this game, if exists. 5

Solution:

3. As per the rule of auction, v_1 and v_2 are not common knowledge. b_1 and b_2 are bid values. Let's find the best response for Player-1:

case 1: $v_1 > v_2$

We have to find b_1 .

Player 1 should ensure $b_1 > v_2$ for winning the auction; and $b_1 \leq v_1$ for having non-negative payoff value.

$$\text{i.e., } v_2 < b_1 \leq v_1 \quad \text{--- ①}$$

case 2: $v_1 < v_2$

For winning the bid and having non-negative payoff, the best response for player 1 is:

$$b_1 = v_1 \quad \text{--- ②}$$

Combining ① and ②, we have

$$\boxed{b_1 = v_1}$$

Similarly, we can reason about the best response of player 2:

$$\boxed{b_2 = v_2}$$

So, the PSNE is $(b_1 = v_1, b_2 = v_2)$.

Evaluation:

- If it is mentioned that v_1 and v_2 are not common knowledge: 1
 - Mention two different cases: 2
 - Combine the solution of two cases and get the correct answer: 2
4. (a) Write down the necessary and sufficient conditions such that a mixed strategy profile $(\sigma_1^*, \dots, \sigma_n^*)$ is a MSNE. 2
- (b) Using the necessary and sufficient conditions for an MSNE, investigate if there exists an MSNE with the *support* $\{A, B\} \times \{X, Y\}$. 3

		P-2	
		X	Y
P-1	A	20, 0	0, 10
	B	0, 90	20, 0

Solution:

4. (a) Necessary and sufficient conditions for a mixed strategy $(\sigma_1^*, \dots, \sigma_n^*)$ to be an MSNE:

(i) $u_i(s_i, \sigma_{-i}^*)$ is the same $\forall s_i \in \mathcal{S}(\sigma_i^*)$

and

(ii) $u_i(s_i, \sigma_{-i}^*) \geq u_i(s_i', \sigma_{-i}^*)$
 $\forall s_i \in \mathcal{S}(\sigma_i^*), \forall s_i' \notin \mathcal{S}(\sigma_i^*)$

Evaluation:

- Correctly written conditions with the explanation of *support*: 2
- Correctly written conditions without the explanation of *support*: 1.5
- For incorrect or partially written conditions: 0

4.(b) The first condition of the necessary and sufficient rule for MSNE applies here, because all the pure strategies are in support (S) according to the question. So, applying the first condition as follows:

For player 1: $p(A) = a$, $p(B) = 1-a$

For player 2: $p(X) = x$, $p(Y) = 1-x$

$$u_1(A) = u_1(B)$$

$$\Rightarrow 20 \cdot x + 0(1-x) = 0 \cdot x + 20(1-x)$$

$$\Rightarrow x = \frac{1}{2}$$

$$\therefore 1-x = \frac{1}{2}$$

$$\text{Also, } u_2(X) = u_2(Y)$$

$$\Rightarrow 0 \cdot a + 90 \cdot (1-a) = 10 \cdot a + 0 \cdot (1-a)$$

$$\Rightarrow a = \frac{9}{10}$$

$$\therefore 1-a = \frac{1}{10}$$

The MSNE is $\left[\left(\frac{9}{10}, \frac{1}{10} \right), \left(\frac{1}{2}, \frac{1}{2} \right) \right]$

Evaluation:

- Why condition-2 is trivially true: 0.5
- For satisfaction of condition-1, calculation method of mixed strategy for two players: 1+1
- Correct answer: 0.5

5. Apply iterative elimination of strongly dominated strategies to the following problem and show the reduced matrix in each iteration. 5

		P-2		
		X	Y	Z
P-1	A	5, 1	2, 2	1, 3
	B	2, 3	5, 4	6, 0
	C	3, 2	3, 0	0, 4

5.

		P-2		
		X	Y	Z
P-1	A	(5, 1)	(2, 2)	(1, 3)
	B	(2, 3)	(5, 4)	(6, 0)
	C	(3, 2)	(3, 0)	(0, 4)

A mix of strategies A and B eliminating strategy C is possible, because P-1's payoff for either A or B or both is/are greater than that of C.

Let, P-1 chooses A and B with probabilities $\frac{1}{2}$ and $\frac{1}{2}$ respectively.

Then strategy C can be eliminated with the above mix, because the expected payoff is higher than playing C for any strategy of player-2.

After eliminating the strongly dominated strategy C we have:

		P-2		
		X	Y	Z
.		(3, 2)	(2, 2)	(1, 3)

P-1	A	(5, 1)	(2, 2)	(1, 3)
	B	(2, 3)	(5, 4)	(6, 0)

Now, strategy X of P2 is strongly dominated by the pure strategy Y. So, we have:

		P-2	
		Y	Z
P-1	A	(2, 2)	(1, 3)
	B	(5, 4)	(6, 0)

Further we apply eliminations using pure strategies and end up with the following matrix:

		P-2	
		Y	
P-1	B	(5, 4)	

Evaluation:

- If the answer is the matrix cannot be reduced: 0.0
 - Mention that mixed strategy of A and B eliminating C: 1
 - A specific instance of mixed strategy of A and B eliminating C: 1
 - For the remaining 3 steps eliminating strategies using pure strategies: $1 \times 3 = 3$
6. Suppose a team of n UAVs (unmanned aerial vehicles) $r_1, r_2, \dots, r_n$ are engaged in surveillance of a workspace W , in a coordinated way. The UAVs are configured (i.e., positioned) in W in a way that they maximize the *coverage area* collectively. Assume that, at some time point t , the UAVs are configured in positions $\{p_1, p_2, \dots, p_n\}$ in a way that they maximally cover W . At time $t + 1$, one of the UAVs r_i is made non-functional somehow, making a partial void in the surveillance task. Now, answer the following:
- An UAV r_k from the remaining $n - 1$ functional UAVs move close to p_i (the position of r_i) to compensate the void. Suggest a technique how the functional UAVs unanimously choose an r_k for this task in a *decentralized way*. 5

6. To solve the problem an auction can be formulated where

Bidders: functional robots

Auction type: decentralized

Common knowledge: Information of failed robot, along with its position.

Bidding value for each robot:

Each robot r_k considers the case in which it is the chosen robot to fulfill the void.

The new location of r_k is a location in the neighbourhood $p'_k = N(p_i)$ of the non-functional robot position p_i such that:

- (i) Travel cost from p_k to p'_k is minimized
- (ii) Coverage is maximized with $n-1$ robots where p'_k is the new position of r_k .

Decentralized approach:

Each robot r_k sends its bidding value to all other $n-2$ robots. This enables to choose the winner robot unanimously.

Evaluation:

-
- Formulation of a decentralized auction: 2
 - Bidding value calculation (with two criteria): 2
 - with a single criteria: 1
 - Mechanism of choosing the winner robot: 1

7. Suppose in a disaster-hit area, a target located at p_d is waiting to be served by a ground robot located at p_r . There are many paths $\{\pi_i\}$ between p_r and p_d . You have to do a tri-criteria search with the following *weighted* criteria:

- (i) Path cost $c(\pi_i)$ of the robot with weight $\alpha_1 \geq 0$,
- (ii) Wait time $\omega(\pi_i)$ of the target with weight $\alpha_2 \geq 0$,
- (iii) A fixed hard deadline D , by which the target must be served, with weight $\alpha_3 \geq 0$

Note that $\alpha_1 + \alpha_2 + \alpha_3 = 1$. Now, answer the following:

- Formulate strict dominance and weak dominance.

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7.

In the given scenario:

Let π_i and π_j be two paths from P_r to P_d

Hard deadline satisfaction for path π_i :

$$I(w_i > D) = \begin{cases} 0 & \text{if } w_i \not> D \\ 1 & \text{if } w_i > D \end{cases}$$

Strict dominance:

π_i strictly dominates π_j if all the following holds:

$$(i) \alpha_1 c(\pi_i) \leq \alpha_1 c(\pi_j)$$

$$\Rightarrow c(\pi_i) \leq c(\pi_j) \quad \text{--- ①}$$

$$(ii) \alpha_2 w(\pi_i) \leq \alpha_2 w(\pi_j)$$

$$\Rightarrow w(\pi_i) \leq w(\pi_j) \quad \text{--- ②}$$

$$(iii) \alpha_3 I(w_i > D) \leq \alpha_3 I(w_j < D)$$

$$\Rightarrow I(w_i > D) \leq I(w_j < D) \quad \text{--- ③}$$

(iv) At least one of ①, ② or ③ holds a strict inequality

Weak dominance:

π_i weakly dominates π_j if ①, ② and ③ hold.

Evaluation:

- Formulation of strong dominance (with path cost and wait time): 1.5
- Formulation of weak dominance (with path cost and wait time): 1.5
- *Deadline* properly defined: 2